

7 decimal

48

2^6	2^5	2^4	2^3	2^2	2^1	2^0
64	32	16	8	4	2	1

7 →

0	0	0	0	1	1	1
---	---	---	---	---	---	---

15 →

0	0	0	1	1	1	1
---	---	---	---	---	---	---

18 →

0	0	1	0	0	1	0
---	---	---	---	---	---	---

55 →

0	1	1	0	1	1	1
---	---	---	---	---	---	---

12
12
24
32

54

14

bitwise

$\&$ $\rightarrow$ set bit to 1 if all bits are 1

$|$ $\rightarrow$ set bit to 1 if at least one bit is 1

$\sim$ $\rightarrow$ reverse bit

$\wedge$ $\rightarrow$ set bit to 1 if bits are different

$<<$ $\rightarrow$ shift bit to left by given number

$>>$ $\rightarrow$ right

7 $\rightarrow$ 0 0 0 0 0 0 1 1 1

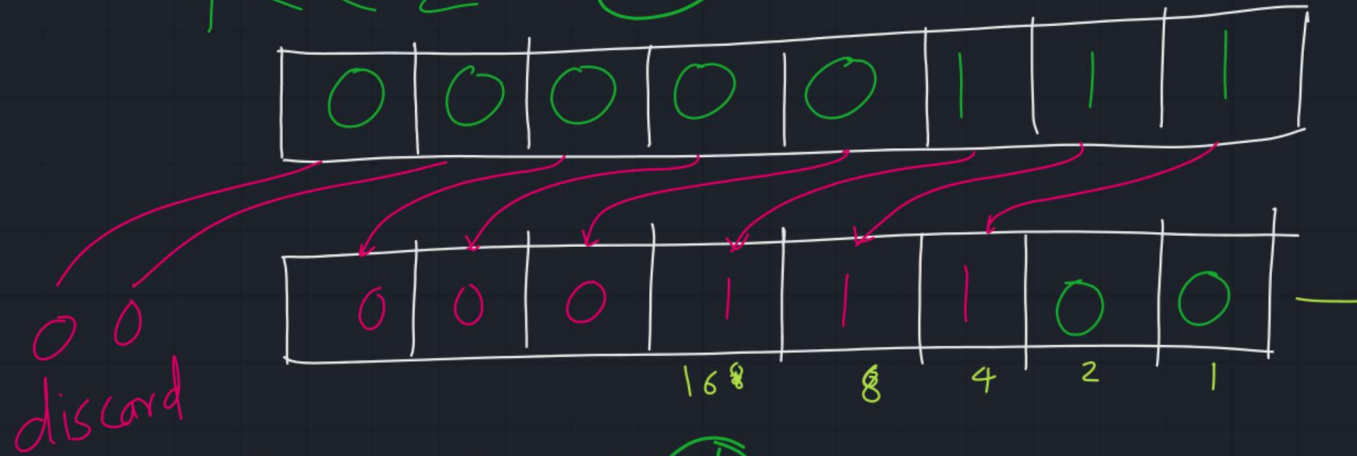
5 $\rightarrow$ 0 0 0 0 0 0 1 0 1

7 & 5 0 0 0 0 0 0 1 0 1 $\rightarrow$ 5

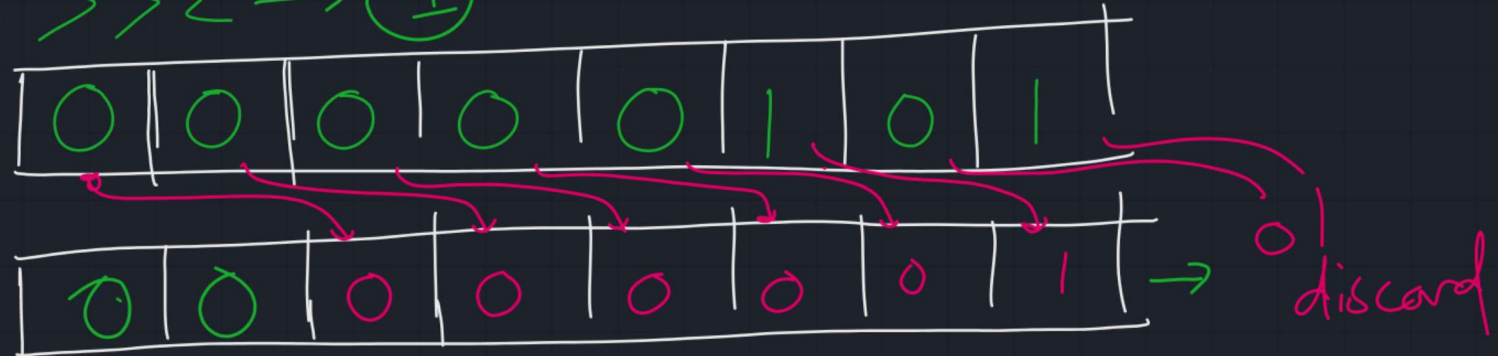
7 | 5 0 0 0 0 0 0 1 1 1 $\rightarrow$ 7

7 ^ 5 0 0 0 0 0 0 0 1 0 $\rightarrow$ 2

$$7 << 2 \rightarrow 28$$



$$5 >> 2 \rightarrow 1$$



$$\ln x = (-x-1) =$$

$$\ln 5 = (-5-1) = -6$$

Unary
→ operators with only one operand

!, ~, ++, --

├→ post ++	├→ post --
└→ pre ++	└→ pre --

x = 10 ← int x = 10;

x = 10 ← x++; (first print the increment for next use)

x = 11 ← s.o.p(x);

$y = 5 \leftarrow \text{int } y = 5;$

$y = 6 \leftarrow ++y;$ (first increment then print)

$y = 6 \leftarrow \text{s.o.p}(y);$

$\text{int } n = 15; \rightarrow n = 15$

$n--; \rightarrow n = 15$

$\text{s.o.p}(n); \rightarrow n = 14$

$--n; \rightarrow n = 13$

$\text{s.o.p}(n); \rightarrow n = 13$

int x = 5;

S.o.p(x++ + ++x);



operator precedence

-
- ()
- ++, -- → ① pre/post inc/dec
- ~, ! → ② unary
- *, /, % → ③ arithmetic
- +, - → ④ arithmetic
- <<, >> → ⑤ Bitwise shift
- <, >, <=, >= → ⑥ Comparison
- !=, == → ⑦ Comparison
- &, |, ^ → ⑧ Bitwise & | ^
- &&, || → ⑨ Logical &&, ||
- ?: → ⑩ Ternary
- =, +=, -= → ⑩ Assignment

very
high
high

low